

help rapid growth of the solar power sector in our country via rooftop and other low-power solar plants. MNRE should quickly clarify when the power utilities will be instructed to buy solar power from 5kW-100kW solar plants at the same rate as +1MW plants. As more and more rooftop solar power plants feed power into the grid, local power utility companies should be happy to buy power at a higher price as this will help them earn carbon credits.

MNRE needs to remove all the remaining roadblocks to encourage rooftop and backyard solar power plants of 5 kW to 500 kW capacity to feed their solar power into the grid and augment shortage of supply from utility companies. Every other country in the world pays a high rate for such solar power fed into the grid and this makes such an investment very rewarding. The available metering technology is secure enough for the government to not worry about misuse.

Major contributors to this report

P.S. Deodhar
Former chairman,
Electronics Commission,
government of India



K. Subramanya
CEO, Tata BP Solar India

Biju P.C.
Manager-sales and
marketing, Emmvee
Photovoltaic Power



Rajesh Jain
Assistant general manager,
Central Electronics

Eligibility criteria for project proponent

While the government is yet to announce the policy for rooftop grid-connected power plants, it has laid down certain guidelines for rooftop PV and other small solar power plants connected to distribution networks at voltage levels below 33 kV. Hereinafter, the programme is referred to as Rooftop PV & Small Solar Power Generation Programme (RPSSGP).

Technical criteria. The project schemes that propose to deploy PV modules and inverter systems are considered to be technically qualified and eligible for participation in the RPSSGP scheme only if these comply with relevant IEC/BIS standards and/or applicable standards as specified by the Central Electricity Authority (CEA). For solar PV projects to be selected under this scheme, it is mandatory that these are based on crystalline silicon technology and use modules manufactured in India. There will be no mandatory domestic content requirement for projects based on other technologies. For solar thermal technology, it is mandatory that the technology is demonstrated and such projects should be operational for one year. Project proponents should submit documentary evidence and an undertaking in this regard along with their applications to the competent authority in the state.

Metering arrangements. Metering arrangements should be made by the project proponents in consultation with the distribution utility keeping in view the guidelines or regulations notified by the respective state electricity regulatory commissions, if any. Meters should comply with the requirements of CEA regulations on the meter installation and operation.

Financial criteria. The project proponents should submit their letters of commitment along with board resolution for equity investments in the project, calculated on the basis of Rs 40 million per megawatt on a pro-rata basis.

Infrastructure criteria for land requirement. The project proponents should make arrangements for land required for the project as per conditions outlined by respective state competent authority.

Infrastructure criteria for grid connectivity requirement. The plant should be designed for interconnection with the grid at the distribution network at the voltage level depending on the installed capacity of the rooftop PV or small solar system generator.

(For the complete requirement list please visit www.mnre.gov.in)

Government subsidies

Government subsidies for standalone rooftop PV installations vary from state to state. For standalone use, today, there is a system that comes with a 40-watt solar panel, 40V battery and two LED bulbs for around Rs 8000, to which the government provides close to a 50 per cent subsidy. By installing this solar system, monthly electricity bills fall by about Rs 60 a month (calculated at the rate of Rs 2 per unit). Conventional electricity bills are currently about Rs 6-7 per unit for residential purposes and Rs 10-12 per unit for commercial establishments.

The government is yet to announce the final specifications and subsidies (or incentives) for rooftop installations that feed into the grid.

Some challenges

The major challenge faced by the project developers is to realise the benefits

of the policy. Not only do the specifications vary from state to state but also the buying rates. Some companies are buying power at as low as Rs 11 per unit, thus discouraging players from venturing into this segment as they make only a marginal profit.

Another challenge is that some unscrupulous players, instead of injecting solar power from their panels, may connect the mains power (conventional electricity) from their neighbour's building to the grid. In such a scenario it becomes difficult for the energy meter to detect the mode of power. However, this challenge is being overcome by introducing innovative DC energy meters that detect the kind of power being transmitted and accept only the power generated by solar PV panels. While this concept is yet to take off in India, there are some specifications laid down by the government with regard to metering. ●

The author is an assistant editor at EYF